Almost rational Diophantine septuples

Problèmes ouverts — A. Dujella · 2026

Problem 3.3 (Dujella, 2026). Are there infinitely many almost rational Diophantine septuples, i.e. sets of seven rationals such that all products increased by 1 are squares, with one exception ?

Remark. Gibbs (2016) and Dujella-Kazalicki-Petričević (2018) found several examples of almost rational Diophantine septuples. For example, in $\{243/560, 1147/5040, 1100/63, 7820/567, 95/112, 38269/6480, 196/$ the only missing condition is that $38269/6480 \cdot 196/45 + 1$ is not a square.